

Mathematics A

Unit 1 • Chapter OL-T31 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

4 classified items • 23 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

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Contents

Unit 1 • Expertise • Unit 1 • Chapter OL-T31 • Expertise Q16+ • 4 items • 23 marks

Chapter	Items	Marks	Starts
01. OL-T31 Coordinate Geometry	4	23	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Coordinate Geometry

Unit 1 • Expertise • 23 marks • 4 item(s)

June 2022 | 4MA1-2H Q22 | 7 marks

Kite or symmetric vertex coordinates

January 2020 | 4MA1-1H Q22 | 6 marks

Coordinates from isosceles and gradient constraints

January 2021 | 4MA1-2H Q18 | 4 marks

Rectangle or parallelogram missing vertex

January 2023 | 1H Q23 | 6 marks

Read a plotted coordinate pair

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-2H Q22 Q22

7 marks

- 22 $ABCD$ is a kite, with diagonals AC and BD , drawn on a centimetre square grid, with a scale of 1 cm for 1 unit on each axis.

A is the point with coordinates $(-3, 4)$

The diagonals of the kite intersect at the point M with coordinates $(0, 2)$

Given that $AB = AD = 6.5$ cm and the x coordinate of B is positive,
find the coordinates of the points B and D .

(..... ,)

(..... ,)

WORKING SPACE

22 Triangle HJK is isosceles with $HJ = HK$ and $JK = \sqrt{80}$

H is the point with coordinates $(-4, 1)$

J is the point with coordinates $(j, 15)$ where $j < 0$

K is the point with coordinates $(6, k)$

M is the midpoint of JK .

The gradient of HM is 2

Find the value of j and the value of k .

$j = \dots\dots\dots$

$k = \dots\dots\dots$

YOUR WORKING

18 A rectangle $ABCD$ is to be drawn on a centimetre grid such that

A has coordinates $(-4, -2)$

B has coordinates $(1, 10)$

C has coordinates $(19, a)$

D has coordinates (b, c)

(a) Work out the value of a , the value of b and the value of c .

$a =$

$b =$

$c =$

(4)

(b) Calculate the perimeter, in centimetres, of rectangle $ABCD$.

..... cm

(3)

YOUR WORKING

23 $ABCD$ is a kite.

$$AB = AD \text{ and } CB = CD$$

The point B has coordinates $(k, 1)$ where k is a negative constant.

The point D has coordinates $(8, 7)$

The straight line L passes through the points B and D

The straight line L is parallel to the line with equation $5y - 3x = 6$

Find an equation of AC

Give your answer in the form $px + qy = r$ where p , q and r are integers.

Show your working clearly.



YOUR WORKING

(Total for Question 23 is 6 marks)

YOUR WORKING